

First Version of the Division Algorithm and Hardware:

Iteration	Step	Quotient	Divisor	Remainder
0	Initial values	0000	0010 0000	0000 0111
1	1: Rem = Rem - Div	0000	0010 0000	⓪110 0111
	2b: Rem < 0 $\Rightarrow$ +Div, sll Q, Q0 = 0	0000	0010 0000	0000 0111
	3: Shift Div right	0000	0001 0000	0000 0111
2	1: Rem = Rem - Div	0000	0001 0000	⓪111 0111
	2b: Rem < 0 $\Rightarrow$ +Div, sll Q, Q0 = 0	0000	0001 0000	0000 0111
	3: Shift Div right	0000	0000 1000	0000 0111
3	1: Rem = Rem - Div	0000	0000 1000	⓪111 1111
	2b: Rem < 0 $\Rightarrow$ +Div, sll Q, Q0 = 0	0000	0000 1000	0000 0111
	3: Shift Div right	0000	0000 0100	0000 0111
4	1: Rem = Rem - Div	0000	0000 0100	⓪000 0011
	2a: Rem $\geq$ 0 $\Rightarrow$ sll Q, Q0 = 1	0001	0000 0100	0000 0011
	3: Shift Div right	0001	0000 0010	0000 0011
5	1: Rem = Rem - Div	0001	0000 0010	⓪000 0001
	2a: Rem $\geq$ 0 $\Rightarrow$ sll Q, Q0 = 1	0011	0000 0010	0000 0001
	3: Shift Div right	0011	0000 0001	0000 0001

Initial Setup

- **Iteration 0:** Initial values are set.
 - **Quotient:** 0000 (initialized to 0)
 - **Divisor:** 0010 0000 (32 in decimal)
 - **Remainder:** 0000 0111 (7 in decimal)

Iteration 1

1. **Step 1:** Subtract the divisor from the remainder.
 - Remainder = Remainder - Divisor = 0000 0111 - 0010 0000 = 1110 0111 (which is negative in two's complement)
2. **Step 2b:** Since the remainder is negative, add the divisor back to the remainder and shift the quotient left, setting the least significant bit (LSB) to 0.
 - Remainder = Remainder + Divisor = 1110 0111 + 0010 0000 = 0000 0111
 - Quotient = 0000 (shifted left, LSB set to 0)
3. **Step 3:** Shift the divisor right by 1.
 - Divisor = 0001 0000 (16 in decimal)

Iteration 2

1. **Step 1:** Subtract the divisor from the remainder.
 - Remainder = $0000\ 0111 - 0001\ 0000 = 1111\ 0111$ (negative)
2. **Step 2b:** Add the divisor back and shift the quotient left, LSB set to 0.
 - Remainder = $1111\ 0111 + 0001\ 0000 = 0000\ 0111$
 - Quotient = 0000
3. **Step 3:** Shift the divisor right by 1.
 - Divisor = 0000 1000 (8 in decimal)

Iteration 3

1. **Step 1:** Subtract the divisor from the remainder.
 - Remainder = $0000\ 0111 - 0000\ 1000 = 1111\ 1111$ (negative)
2. **Step 2b:** Add the divisor back and shift the quotient left, LSB set to 0.
 - Remainder = $1111\ 1111 + 0000\ 1000 = 0000\ 0111$
 - Quotient = 0000
3. **Step 3:** Shift the divisor right by 1.
 - Divisor = 0000 0100 (4 in decimal)

Iteration 4

1. **Step 1:** Subtract the divisor from the remainder.
 - Remainder = $0000\ 0111 - 0000\ 0100 = 0000\ 0011$ (positive)
2. **Step 2a:** Since the remainder is non-negative, shift the quotient left and set the LSB to 1.
 - Quotient = 0001
3. **Step 3:** Shift the divisor right by 1.
 - Divisor = 0000 0010 (2 in decimal)

Iteration 5

1. **Step 1:** Subtract the divisor from the remainder.
 - Remainder = $0000\ 0011 - 0000\ 0010 = 0000\ 0001$ (positive)
2. **Step 2a:** Shift the quotient left and set the LSB to 1.
 - Quotient = 0011
3. **Step 3:** Shift the divisor right by 1.
 - Divisor = 0000 0001 (1 in decimal)

Final Result

- The final quotient is 0011 (3 in decimal).
- The final remainder is 0000 0001 (1 in decimal).